Arithmetic Sequences as Linear Functions

Answer Key

Algebra 1

Quick Answers

1. 39	4. 93	7. 183	10. 645
2. 23	5. 83	8. 8	
3. 580	6. 15	9. 21	

Common wrong answers

2. *If they got 103: added the common difference instead of subtracting it*
3. *If they got 665: multiplied by n instead of n - 1 (counted hour 1 as a step)*
7. *If they got 192: added 9 cm for all 12 chairs instead of for the 11 stacked on top*

1. Tile pattern 4, 11, 18, 25, ... — recursive formula and stage 6.

Step 1 (find d): subtract consecutive terms — $11 - 4 = 7$, $18 - 11 = 7$, $25 - 18 = 7$. The difference is constant, so the sequence is arithmetic with $d = 7$. Step 2 (write both lines): $a_1 = 4$ and $a_n = a_{n-1} + 7$ for $n \geq 2$. Step 3 (reach stage 6): from stage 1 to stage 6 there are $6 - 1 = 5$ steps, so $a_6 = 4 + 5(7) = \boxed{39}$ tiles. (By hand: 4, 11, 18, 25, 32, 39.)

2. Drone heights 63, 58, 53, ... — recursive formula and reading 9.

Step 1 (find d): $58 - 63 = -5$ and $53 - 58 = -5$, so the common difference is negative: $d = -5$. A falling sequence still *adds* d; d itself is negative. Step 2 (write both lines): $a_1 = 63$ and $a_n = a_{n-1} - 5$ for $n \geq 2$. Step 3 (reach reading 9): $9 - 1 = 8$ steps, so $a_9 = 63 + 8(-5) = 63 - 40 = \boxed{23}$ meters.

3. Tank: 240 L at the end of hour 1, +85 L each later hour — liters at the end of hour 5.

Step 1 (identify the givens with units): $a_1 = 240$ liters (end of hour 1) and $d = +85$ liters per hour. Step 2 (write both lines): $a_1 = 240$ and $a_n = a_{n-1} + 85$ for $n \geq 2$, where a_n is in liters. Step 3 (reach hour 5): hour 1 is the starting value, so hours 2, 3, 4, 5 are $5 - 1 = 4$ steps: $a_5 = 240 + 4(85) = 240 + 340 = \boxed{580 \text{ L}}$.

4. $a_1 = 12$, $a_n = a_{n-1} + 9$ — read the rule, then find a_{10}.

Step 1 (say the rule in words): start at 12, and to get any term add 9 to the term before it. Step 2 (count the steps): reaching a_{10} from a_1 takes $10 - 1 = 9$ additions of 9, not 10. Step 3 (compute): $a_{10} = 12 + 9(9) = 12 + 81 = \boxed{93}$.

5. $a_1 = 7$, $a_n = a_{n-1} + 4$ **— explicit rule and** a_{20}**.**
Step 1 (translate the pieces): the common difference $d = 4$ is the slope m, because each step of 1 in n changes the value by 4. Step 2 (find the intercept): $a_n = 7 + (n-1)4 = 4n+3$, so $m = 4$ and $b = 3$. Here b is the value the line would take at $n = 0$ — one step *before* the first term, not the first term itself. Step 3 (evaluate): $a_{20} = 4(20) + 3 = \boxed{83}$, which matches $7 + 19(4) = 83$.

6. $a_1 = 5$, $a_n = a_{n-1} + 6$ **— which term equals 89?**
Step 1 (set up an equation instead of listing): the explicit form of the recursive rule is $a_n = 5+(n-1)6$, and we need that to equal 89. Step 2 (solve): $5+6(n-1) = 89 \Rightarrow 6(n-1) = 84 \Rightarrow n-1 = 14$. Step 3: $n = \boxed{15}$, so the 15th term is 89. (Check: $5 + 14(6) = 89$.)

7. Chair stack: 84 cm for one chair, +9 cm per added chair — height of 12 chairs.
Step 1 (identify the givens with units): $a_1 = 84$ centimeters for the first chair, and each chair after that adds $d = 9$ centimeters. Step 2 (write both lines): $a_1 = 84$ and $a_n = a_{n-1} + 9$ for $n \geq 2$, with a_n in centimeters. Step 3 (reach 12 chairs): 12 chairs means the first chair plus $12 - 1 = 11$ added chairs: $a_{12} = 84 + 11(9) = 84 + 99 = \boxed{183 \text{ cm}}$.

8. Table with $a_3 = 20$ **and** $a_7 = 44$ **— recursive formula and** a_1**.**
Step 1 (count the steps between the known terms): going from $n = 3$ to $n = 7$ takes $7-3 = 4$ equal steps, and the total change is $44 - 20 = 24$. Step 2 (find d): $d = 24 \div 4 = 6$, so the recursive rule is $a_n = a_{n-1} + 6$. Step 3 (walk backwards to a_1): from a_3 back to a_1 is 2 steps *down*: $a_1 = 20 - 2(6) = \boxed{8}$. (Check forward: 8, 14, 20, 26, 32, 38, 44.)

9. Savings: 45 dollars at end of week 1, +18 dollars weekly — first week the balance reaches 405 dollars.
Step 1 (write both lines): $a_1 = 45$ dollars and $a_n = a_{n-1} + 18$ for $n \geq 2$, with a_n the balance in dollars at the end of week n. Step 2 (set up the equation): the explicit form is $a_n = 45 + (n-1)18$; set it equal to 405. Step 3 (solve): $18(n-1) = 360 \Rightarrow n - 1 = 20$, so $n = \boxed{21}$ — the balance first reaches 405 dollars at the end of week 21.

10. Theater: row 1 has 22 seats, each later row has 3 more — total seats in the first 15 rows.
Step 1 (recursive rule): $a_1 = 22$ seats and $a_n = a_{n-1} + 3$ for $n \geq 2$. (ii) A recursive rule gives one row at a time, so to total 15 rows you would have to generate all 15 terms first — the rule describes the *step*, not the *sum*. Step 2 (list what you need): the last row is $a_{15} = 22 + 14(3) = 64$ seats. Step 3 (add the arithmetic series by pairing first and last): $S_{15} = \dfrac{15(22+64)}{2} = \dfrac{15 \cdot 86}{2} = \boxed{645}$ seats in the first 15 rows.